

Physics 828: Homework Set No. 6

**Due date: Friday, February 24, 2011, 1:00pm
in PRB M2043 (Biao Huang's office)**

Total point value of set: 60 points

Problem 1 (5 pts.): Exercise 13.1.4 (Shankar, p. 357)

Problem 2 (15 pts.): Exercise 13.1.5 (Shankar, p. 359)

Problem 3 (5 pts.): Exercise 13.3.1 (Shankar, p. 366)

Problem 4 (5 pts.): Exercise 13.3.2 (Shankar, p. 366)

Problem 5 (10 pts.): Exercise 13.4.1 (Shankar, p. 371)

Problem 6 (10 pts.): Exercise 13.4.2 (Shankar, p. 371)

Problem 7 (5 pts.): Exercise 14.3.3 (Shankar, p. 384)

Problem 8 (5 pts.): Exercise 14.3.5 (Shankar, p. 385)

Exercise 13.1.4 (357)

Eq. (12.6.19) states for the Coulomb potential the following asymptotic behavior for bound states:

$$U_E(r) \sim r^{\frac{me^2}{\hbar^2}} e^{-\kappa r} \quad \text{where } \kappa = \sqrt{\frac{-2mE}{\hbar^2}}$$

Working this out for the eigen-energies

$$E_n = -\frac{me^4}{2\hbar^2 n^2} \rightarrow \kappa_n = \sqrt{\frac{m^2 e^4}{\hbar^4 n^2}} = \frac{1}{n} \frac{me^2}{\hbar^2} = \frac{1}{na_0}$$

we find

$$R_n(r) = \frac{U_n(r)}{r} \sim \frac{1}{r} r^n e^{-r/na_0} = r^{n-1} e^{-r/(na_0)}$$

as found in Eq. (13.1.26) ✓

Exercise 13.1.5 (p. 359)

$$|\psi_{nlm}(t)\rangle = e^{-\frac{i}{\hbar} \hat{H} t} |nlm\rangle = e^{-\frac{i}{\hbar} E_n t} |nlm\rangle$$

$$\Rightarrow \langle \psi_{nlm}(t) | \hat{\Omega} | \psi_{nlm}(t) \rangle = \langle nlm | e^{\frac{i}{\hbar} E_n t} \hat{\Omega} e^{-\frac{i}{\hbar} E_n t} | nlm \rangle$$

$$= \langle nlm | \hat{\Omega} | nlm \rangle \text{ independent of } t$$

$$\Rightarrow \frac{d}{dt} \langle \psi_{nlm}(t) | \hat{\Omega} | \psi_{nlm}(t) \rangle \stackrel{\text{Ehrenfest}}{=} \langle \psi_{nlm}(t) | \frac{i}{\hbar} [\hat{H}, \hat{\Omega}] | \psi_{nlm}(t) \rangle = 0$$

$$\text{For } \hat{\Omega} = \hat{\vec{R}} \cdot \hat{\vec{P}}, [\hat{H}, \hat{\Omega}] = \left[\frac{\hat{\vec{P}}^2}{2m} - \frac{e^2}{|\hat{\vec{R}}|}, \hat{\vec{R}} \cdot \hat{\vec{P}} \right] =$$

$$= \frac{1}{2m} [\hat{P}_i \hat{P}_i, \hat{R}_j \hat{P}_j] - e^2 \left[\frac{1}{\sqrt{\hat{R}_i \hat{R}_i}}, \hat{R}_j \hat{P}_j \right]$$

$$= \frac{1}{2m} (\hat{P}_i [\hat{P}_i, \hat{R}_j \hat{P}_j] + [\hat{P}_i, \hat{R}_j \hat{P}_j] \hat{P}_i)$$

$$- e^2 \left(\underbrace{\left[\frac{1}{\sqrt{\hat{R}_i \hat{R}_i}}, \hat{R}_j \right] \hat{P}_j}_0 + \hat{R}_j \left[\frac{1}{\sqrt{\hat{R}_i \hat{R}_i}}, \hat{P}_j \right] \right)$$

$$= \frac{1}{2m} \left(\hat{P}_i \underbrace{[\hat{P}_i, \hat{R}_j]}_{i\hbar \delta_{ij}} \hat{P}_j + \hat{P}_i \hat{R}_j \underbrace{[\hat{P}_i, \hat{P}_j]}_0 + \underbrace{[\hat{P}_i, \hat{R}_j]}_{i\hbar \delta_{ij}} \hat{P}_j \hat{P}_i + \hat{R}_j \underbrace{[\hat{P}_i, \hat{P}_j]}_0 \right)$$

$$- e^2 \hat{R}_j \left[\frac{1}{\sqrt{\hat{R}_i \hat{R}_i}}, \hat{P}_j \right]$$

$$= \frac{i\hbar}{m} \hat{\vec{P}}^2 - e^2 \hat{R}_j \left[\frac{1}{\sqrt{\hat{R}_i \hat{R}_i}}, \hat{P}_j \right] = 2i\hbar \underbrace{\frac{\hat{\vec{P}}^2}{2m}}_{\hat{H}} - e^2 \hat{R}_j \left[\frac{1}{|\hat{\vec{R}}|}, \hat{P}_j \right]$$

To work out the last commutator, we work out all its matrix elements in the $\hat{\mathbf{R}}$ -eigenbasis where the action of $\frac{1}{|\hat{\mathbf{R}}|}$ can be evaluated easily:

$$\langle \vec{r} | \hat{R}_j \left[\frac{1}{|\hat{\mathbf{R}}|}, \hat{P}_j \right] | \vec{r}' \rangle = \quad (\vec{r} = (x_1, x_2, x_3))$$

$$= \langle \vec{r} | \frac{\hat{R}_j}{|\hat{\mathbf{R}}|} \hat{P}_j | \vec{r}' \rangle - \langle \vec{r} | \hat{R}_j \hat{P}_j \frac{1}{|\hat{\mathbf{R}}|} | \vec{r}' \rangle$$

$$= \langle \vec{r} | \frac{x_j}{|\hat{\mathbf{R}}|} \hat{P}_j | \vec{r}' \rangle - \langle \vec{r} | x_j \hat{P}_j \frac{1}{|\hat{\mathbf{R}}|} | \vec{r}' \rangle$$

$$(r \equiv |\vec{r}|) = \frac{x_j}{r} \underbrace{\langle \vec{r} | \hat{P}_j | \vec{r}' \rangle}_{-i\hbar \delta(\vec{r}-\vec{r}') \frac{\partial}{\partial x_j}} - r_j \underbrace{\langle \vec{r} | \hat{P}_j | \vec{r}' \rangle}_{-i\hbar \delta(\vec{r}-\vec{r}') \frac{\partial}{\partial x_j}} \frac{1}{r'}$$

$$= +i\hbar x_j \delta(\vec{r}-\vec{r}') \left(\frac{\partial}{\partial x_j} \frac{1}{r'} - \frac{1}{r'} \frac{\partial}{\partial x_j} \right)$$

Note: in the 1st term $\frac{\partial}{\partial x_j}$ acts on everything that follows!

$$= i\hbar x_j \delta(\vec{r}-\vec{r}') \left(-\frac{x_j'}{(r')^3} + \frac{1}{r'} \frac{\partial}{\partial x_j} - \frac{1}{r'} \frac{\partial}{\partial x_j} \right)$$

can replace $\delta(\vec{r}-\vec{r}') \frac{1}{r} = \frac{\delta(\vec{r}-\vec{r}')}{r'}$

$$= -i\hbar \frac{\vec{r} \cdot \vec{r}'}{|\vec{r}|^3} \delta(\vec{r}-\vec{r}') = -i\hbar \frac{\vec{r} \cdot \vec{r}}{r^3} \delta(\vec{r}-\vec{r}') =$$

$$= -i\hbar \frac{1}{r} \delta(\vec{r}-\vec{r}') = -i\hbar \langle \vec{r} | \frac{1}{|\hat{\mathbf{R}}|} | \vec{r}' \rangle$$

This holds for all $|\vec{r}\rangle, |\vec{r}'\rangle \Rightarrow$

$$\hat{R}_j \left[\frac{1}{|\hat{\mathbf{R}}|}, \hat{P}_j \right] = -i\hbar \frac{1}{|\hat{\mathbf{R}}|}$$

Plug this back into the commutator

$$\frac{i}{\hbar} [\hat{H}, \hat{\vec{R}} \cdot \hat{\vec{P}}] = -2\hat{T} - \frac{e^2}{|\hat{\vec{R}}|} = -2\hat{T} + \hat{V}$$

$$\text{From } \langle \psi_{nlm}(t) | \frac{i}{\hbar} [\hat{H}, \hat{\vec{R}} \cdot \hat{\vec{P}}] | \psi_{nlm}(t) \rangle = 0$$

it thus follows that

$$\langle \psi_{nlm}(t) | \hat{T} | \psi_{nlm}(t) \rangle = -\frac{1}{2} \langle \psi_{nlm}(t) | \hat{V} | \psi_{nlm}(t) \rangle$$

$$\text{or } \boxed{\langle \hat{T} \rangle = -\frac{1}{2} \langle \hat{V} \rangle} \quad \forall(t) \text{ in any state } |nlm\rangle$$

$\Rightarrow$ quantum analogue of the virial theorem $\checkmark$

Exercise 13.3.1 (p. 366)

According to the discussion in Example 9.5.1 on page 246 on the nuclear force, the range of the nuclear force is given by the Compton wavelength of the exchange quanta mediating the force:

$$\Delta t \approx \frac{\hbar}{\Delta E} \approx \frac{\hbar}{\mu c^2} \quad \text{and} \quad r = \text{range} = c \Delta t = \frac{\hbar}{\mu c} = \lambda$$

So the problem states that for the pion $\lambda_\pi \sim 1 \text{ fm}$

$$\Rightarrow \underbrace{m_\pi c^2}_{\text{}} = \frac{\hbar c}{\lambda_\pi} \sim \frac{200 \text{ MeV fm}}{1 \text{ fm}} = \underbrace{200 \text{ MeV}}$$

The measured value is $m_\pi c^2 = 138 \text{ MeV}$.

Exercise 13.3.2 (p. 366)

The de Broglie wavelength is defined as

$$\lambda = 2\pi \hbar / p$$

An electron of kinetic energy $\frac{p^2}{2m} = T = 200 \text{ eV}$

has momentum $p = \sqrt{2m \cdot T}$

$$\begin{aligned} \Rightarrow \underbrace{\lambda}_{\text{}} &= 2\pi \frac{\hbar c}{pc} = \frac{(2\pi)(\hbar c)}{\sqrt{2(mc^2)T}} = \frac{2\pi \cdot 200 \text{ MeV fm}}{\sqrt{2 \cdot (0.5 \text{ MeV}) \cdot 2 \times 10^{-4} \text{ MeV}}} \\ &= \frac{\pi \cdot 400 \text{ MeV fm}}{\sqrt{2} \cdot 10^{-2} \text{ MeV}} = 90,000 \text{ fm} = \underbrace{0.9 \text{ \AA}} \end{aligned}$$

Exercise 13.4.1 (p. 371)

Ignoring electron-electron interactions, the Hamiltonian for each of the electrons in an atom with nuclear charge Ze is

$$\hat{H} = \frac{\hat{p}^2}{2m} - \frac{Ze^2}{|\vec{R}|}$$

The replacement $e^2 \rightarrow Ze^2$ leads in Eqs. (13.1.14/16) to the replacement

$$E_n \rightarrow -\frac{m(Ze^2)^2}{2\hbar^2 n^2}$$

Hence $E_n \sim Z^2 \checkmark$

This happens because, at fixed r , the potential increases by a factor Z , but at the same time the mean orbital radii $\langle r \rangle_n \sim n^2 a_0^{\text{eff}} = n^2 \frac{\hbar^2}{m(Ze^2)}$ decrease by a factor Z , so

$$\begin{aligned} E_n &= \frac{1}{2} \left\langle \left(-\frac{Ze^2}{r} \right) \right\rangle_n = -\frac{1}{2} (Ze^2) \left\langle \frac{1}{r} \right\rangle_n = -\frac{1}{2} (Ze^2) \frac{m(Ze^2)}{\hbar^2 n^2} \\ &= -\frac{m(Ze^2)^2}{2\hbar^2 n^2} \end{aligned}$$

Exercise 13.4.2 (p. 371)

$^{238}_{92}\text{U}$ has 92 electrons. The size of the atom is controlled by the expectation value of the orbital radius of the electron with the largest energy. Without electron-electron interactions and relativistic corrections, $\langle r \rangle_n$ depends only on the main quantum number n :

$$\langle r \rangle_n \sim n^2 a_0^{\text{eff}} = n^2 \frac{a_0}{Z} \quad (\text{from exercise 13.4.1})$$

So to answer the question we only need to figure out the main quantum number n of the highest occupied electron orbit.

Including electron spin, each energy eigenstate E_n is $2n^2$ -fold degenerate, i.e. $2n^2$ electrons can occupy orbits with the same $\langle r \rangle_n$.

		sum
$n=1$	2	2
$n=2$	8	10
$n=3$	18	28
$n=4$	32	60
$n=5$	50	110

*) $\langle r \rangle_5 \sim 25 \frac{a_0}{Z_{\text{eff}}}$

For the last electron $Z_{\text{eff}} \sim 1$, and thus

$\langle r \rangle_n \sim 25 \langle r \rangle_H$

So 60 electrons can be put into the states $n \leq 4$, and the last 32 electrons will occupy states with $n=5$ and

$$\langle r \rangle_5 = 25 \cdot \frac{a_0}{92} = 0.27 a_0 = 0.27 \langle r \rangle_{\text{hydrogen}}$$

This is only about the quarter of the hydrogen radius! Obviously, this calculation is wrong by ignoring electron screening of the nuclear charge: *)

Exercise 14.3.3 (p. 384)

given: $[\sigma_i, \sigma_j] = 0$ for $i+j \Rightarrow \sigma_i \sigma_j = -\sigma_j \sigma_i$ ($i+j$)

$$[\sigma_i, \sigma_j] = 2i \varepsilon_{ijk} \sigma_k \Rightarrow \begin{cases} \sigma_z = -i \sigma_x \sigma_y \\ \sigma_y = -i \sigma_z \sigma_x \\ \sigma_x = -i \sigma_y \sigma_z \end{cases}$$

Desired: $\text{Tr}(\sigma_k) = 0$

$$\text{Tr}(\sigma_k) = -i \text{Tr}(\sigma_l \sigma_m) \quad \text{where } l = k+1, m = k+2 \text{ (modulo 3)}$$

$$\Rightarrow m \neq l$$

$$= -i \text{Tr}\left(\frac{1}{2}(\sigma_l \sigma_m - \sigma_m \sigma_l)\right)$$

$$= -\frac{i}{2} [\text{Tr}(\sigma_l \sigma_m) - \text{Tr}(\sigma_m \sigma_l)] = 0 \quad \text{since } \text{Tr}(AB) = \text{Tr}(BA)$$

$$\text{Tr}(\sigma_k) = 0$$

Exercise 14.3.5 (p. 385)

$$M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \sum_{\alpha=0}^3 c_{\alpha} \sigma_{\alpha}$$

(14.3.43)

$$\Rightarrow c_{\alpha} = \frac{1}{2} \text{Tr}(M \sigma_{\alpha})$$

$$c_0 = \frac{1}{2} \text{Tr}\left(\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = \frac{\alpha + \delta}{2}$$

$$c_1 = \frac{1}{2} \text{Tr}\left(\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right) = \frac{1}{2} \text{Tr}\begin{pmatrix} \beta & \alpha \\ \delta & \gamma \end{pmatrix} = \frac{\beta + \gamma}{2}$$

$$c_2 = \frac{1}{2} \text{Tr}\left(\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\right) = \frac{i}{2} \text{Tr}\begin{pmatrix} \beta & -\alpha \\ \delta & -\gamma \end{pmatrix} = \frac{i}{2} (\beta - \gamma)$$

$$c_3 = \frac{1}{2} \text{Tr}\left(\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right) = \frac{\alpha - \delta}{2}$$

$$\Rightarrow M = \frac{1}{2} \left((\alpha + \delta) \mathbb{1} + (\beta + \gamma) \sigma_1 + i(\beta - \gamma) \sigma_2 + (\alpha - \delta) \sigma_3 \right)$$

check: $= \frac{1}{2} \begin{pmatrix} \alpha + \delta + \alpha - \delta & \beta + \gamma + \beta - \gamma \\ \beta + \gamma - (\beta - \gamma) & \alpha + \delta - (\alpha - \delta) \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2\alpha & 2\beta \\ 2\gamma & 2\delta \end{pmatrix} \checkmark$